Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	LTVIP2026TMIDS73502
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

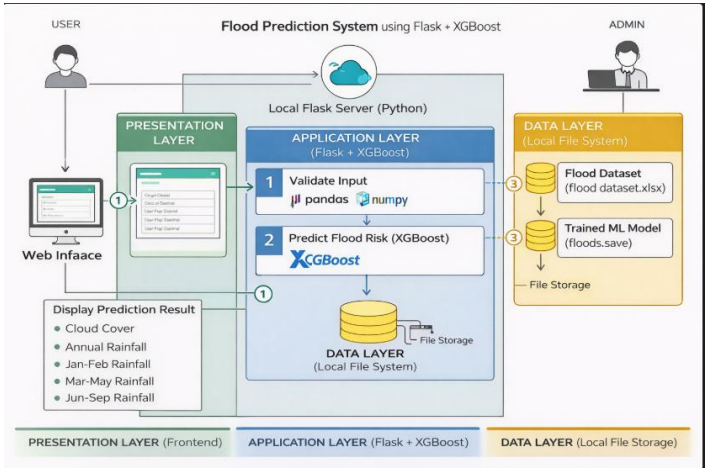

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for data input and result display	HTML, CSS
2.	Application Logic-1	Handles routing and request processing	Python Flask
3.	Application Logic-2	Input validation and DataFrame creation	Pandas, NumPy
4.	Application Logic-3	Flood prediction logic	XGBoost (XGBClassifier)
5.	Database	Historical rainfall dataset used for training	Excel (.xlsx)
6.	Model Storage	Stores trained ML model	Joblib (.save file)
7.	File Storage	Stores dataset and trained model locally	Local File System
8.	Machine Learning Model	Binary classification of severe flood risk	XGBoost Classifier
9.	Infrastructure	Local deployment via Flask server	Localhost (127.0.0.1:5000)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Backend and ML frameworks	Flask, Scikit-learn, XGBoost, Pandas
2.	Security Implementations	Input validation and restricted file access	Flask validation, Python error handling
3.	Scalable Architecture	Modular structure (UI + Backend + ML Model) allows future extension	3-Layer Architecture
4.	Availability	Available whenever Flask server is running locally	Local Server
5.	Performance	Prediction generated within seconds; lightweight model	XGBoost optimized inference

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d